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1. Orthology.2

Orthologs are homologous genes in different organisms that encode proteins with the same function and which have evolved by direct vertical descent..

Orthologs are genes in different species that evolved from a common ancestral gene by speciation, and, in general, **orthologs** retain the same function during the course of evolution. Identification of **orthologs** is a critical process for reliable prediction of gene function in newly sequenced genomes.

2. Write principle of spotted DNA array... 3 marks.

Spotted DNA arrays are produced by printing DNA samples on treated microscope slides. It is made by transferring or spotting DNA clones or PCR product individually onto a solid support where they are immobilized.

3. Write down the main features of Real-time PCR and multiplex PCR? 5

RT-PCR:

Reverse transcription followed by the PCR (RTPCR) leads to the amplification of RNA sequences in cDNA form □ RT-PCR is a sensitive means for detecting, quantifying and cloning specific cDNA molecules. Major screening strategies involve 1). Genetic methods 2). Sequence-dependent screening 3). Screening expression libraries

The main advantage of **real-time PCR** over **PCR** is that **real-time PCR** allows you to determine the initial number of copies of template DNA (the amplification target sequence) with accuracy and high sensitivity over a wide dynamic range.

Multiplex PCR:

Multiplex PCR is used to amplify several different DNA sequences simultaneously.

4. PCR, types and applications? 10

PCR: Polymerase chain reaction, or PCR, is a laboratory technique used to make multiple copies of a segment of DNA. PCR is very precise and can be used to amplify, or copy, a specific DNA target from a mixture of DNA molecules.

Types: Nested PCR, RT-PCR, qPCR, Multiplex PCR.

Applications of PCR

PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc.

PCR-Gene cloning and expression

PCR has been used in gene cloning and screening of genomic libraries.

PCR-Medicine:

PCR has major impact on medicine especially in the field of clinical microbiology/diagnosis Molecular tools have also allowed performing prenatal genetic diagnosis.

PCR-Forensic sciences:

Forensic science is the application of scientific procedures to solve criminal & legal matters Molecular methods are used to establish the filiations of a person or to obtain evidence from minimal samples of saliva, semen or other tissues.

PCR-DNA profiling: DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA

PCR-Agricultural sciences and environment

PCR has also facilitated research in detection of pathogens in plants, animals and environment.

PCR-Molecular paleontology

Molecular paleontology refers to the recovery and analysis of DNA and protein from ancient human, animal and plant remains.

5. Applications of PCR? 3

PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc.

6. DNA sequencing: Benefits and Applications 5

DNA Sequence: DNA sequence information is a prerequisite for planning any substantial manipulation of the DNA □ Information is useful to various fields i.e. molecular and evolutionary biology, metagenomics, medicine, forensics etc. Techniques for DNA sequencing became available in the late 1970s .

DNA sequencing is the process of determining the precise order of nucleotides within a DNA molecule. It includes any method or technology that is used to determine the order of the four bases—adenine, guanine, cytosine, and thymine—in a strand of DNA. The advent of rapid DNA sequencing methods has greatly accelerated biological and medical research and discovery.

7. type of Resistance observed in p1258 plasmid of staphylococcus (2)?

Expression of *Staphylococcus* plasmid genes in *E. coli*□

pSC101 contain Tetracycline resistance

P1248 from Staphylococcus

8. What is epitope? 2

the part of an antigen molecule to which an antibody attaches itself

a molecular region on the surface of an antigen capable of eliciting an immune response and of combining with the specific antibody produced by such a response.

Immunochemical screening involves the use of antibodies that specifically recognize antigenic determinants on the polypeptides synthesized by a target clone

- The molecular target for recognition is generally an epitope

9. Why taq DNA polymerase use in PCR? 2

Taq polymerase is a thermostable DNA polymerase isolated from thermophilic bacterium Thermus aquaticus and is used in PCR to amply DNA fragments.

10. What is the strategy used to synthesis the second cDNA from first strand of cloning? 3

In nested PCR, two sets of primers are used in two successive PCRs

First set of primers allows a **first amplification**

Product of first PCR is subjected to a second PCR using the **second set of primers**.

11. Describe the importance of plasmid copy number and stability (3 Marks)

The effect of plasmid copy number

The number of ribosomes in a cell far exceeds any one class of mRNA. One way of increasing the expression of a cloned gene is to increase the number of the corresponding transcript

Plasmid stability:

Having maximized the expression of a particular gene it is important to consider what effects this will have on the bacterium harboring the recombinant plasmid. The loss of plasmids due to defective partitioning is called segregative instability. Structural instability of plasmids may arise by deletions or rearrangements of DNA.

12. Mutagenesis and its types?10.

Mutagenesis:

Mutagenesis is a process to change the genetic information of an organism. It can occur naturally, or as a result of exposure to mutagens or induced experimentally in laboratory.

Three different methods of site-directed mutagenesis have been devised

i). Cassette mutagenesis

ii). Primer extension

iii). Procedures based on PCR

Cassette mutagenesis:

A synthetic DNA fragment containing the desired mutant sequence is used to replace the corresponding sequence in the wild type gene.

Primer extension: the single primer method

It involves priming in vitro DNA synthesis with a chemically synthesized oligonucleotide (7-20 bp) that carries a base mismatch with the complementary sequence. After *E. coli* transformation, heteroduplex that are formed are either that of original wild type DNA or that containing the mutated base.

PCR methods for site-directed mutagenesis:

Single bases mismatched between the amplification primer and the template become incorporated into the template sequence as a result of amplification. In megaprimer method, the product of the first PCR is used for the second PCR.The advantage of a PCR-based mutagenic protocol is that the desired mutation is obtained with 100% efficiency.

13.Disadvantage of PCR?

Setting up running required high technical skills.

High equipment cost.

High Test Cost

High sterile environment should be provided. (DNA contamination)

14.Four Step of Cloning Strategy?

Any cell based cloning procedure has four essential parts

i). A method of generating DNA fragment for cloning

ii). A reaction that inserts that fragment into the chosen cloning vectors

iii). A means for introducing that recombinant vector into a host cell wherein it is replicated

iv). A method of selecting recipient cells that have acquired the recombinant

15.Applications of microarrays

Microbial gene expression analysis

The predominate application of DNA microarray has been to measure gene expression levels

Profiling in human disease

Arrays have been used to investigate transcriptional profiles associated with human disease and to identify novel disease markers and potential new drug targets

Genotyping

Microarrays have been widely used as single nucleotide polymorphism (SNP) genotyping platforms

16. How polymerase work?

The DNA polymerases are enzymes that create DNA molecules by assembling nucleotides, the building blocks of DNA. ... When creating DNA, DNA polymerase can add free nucleotides only to the 3' end of the newly forming strand. This results in elongation of the newly forming strand in a 5'-3' direction.

17.siRNA technology

Small interfering RNA (siRNA) is the most commonly used RNA interference (RNAi) tool for inducing short- term silencing of protein coding genes. siRNA is a synthetic RNA duplex designed to specifically target a particular mRNA for degradation

18.Applications of siRNA

siRNA) is the method of choice to target specific genes for silencing and has provided immense potential as therapeutic tools

19.Basic PCR reaction:

PCR is a technology in molecular biology used to amplify a single or a few copies of a target DNA across several orders of magnitude, generating millions of copies of a particular DNA sequence.

PCR can also be used to generate DNA fragments for gene cloning by using primers with specific restriction site PCR is a technology in molecular biology used to amplify a single or a few copies of a target DNA across several orders of magnitude, generating millions of copies of a particular DNA sequence

▪ PCR amplifies a specific target region of DNA that may range from a few to several kilo base pairs

▪Basic reaction requires several components

▪ PCR cycles may consist of several steps of denaturation, annealing and extension.

▪ DNA polymerases are enzymes that synthesize DNA molecules by adds nucleotide, in building blocks of DNA

20. Phasmid vectors:

▪ Phasmid vectors are combination of plasmid and ʎ phage sequences ▪ It consists of a plasmid vector carrying a ʎ attachment (ʎatt) site

21. Bacterial artificial chromosomes (BACs):

Bacterial artificial chromosomes are simply plasmids designed for the cloning of very long segments of DNA (100,000 to 300,000 bp)

13. How animal restore their lost function?

Screening by functional complementation

Functional complementation is the process by which a particular DNA sequence compensates for a missing function in a mutant cell and thus restores the wild type phenotype.

Functional complementation is also possible in transgenic animals and plants.

It has been used for complementation in transgenic mice to isolate the Shaker-2 gene.

22. Yeast and Prokaryotic cDNA library?

cDNA libraries are used to express eukaryotic genes in prokaryotes since it does not include introns, and therefore, can be expressed in prokaryotic cells. cDNA libraries remove the large numbers of non-coding regions from the library, and it is also useful for subsequently isolating the gene that codes for that mRNA.

23. Plasmid stability and copy number?

Plasmid copy number and plasmid stability. Many expression systems in research and industry use plasmids as vectors for the production of recombinant proteins or non-proteinous recombinant substances. ... Related factors are plasmid copy number, structural plasmid stability and segregational plasmid stability.

24. siRNA Technology?

siRNA technology. RNA interference (RNAi) represents a mechanism invented by nature to protect the genome. ... The underlying molecular mechanism of gene silencing provides us with short interfering RNAs (siRNAs) which allows to target any gene with high specificity and efficiency

25. what is homologous?

having the same relation, relative position, or structure (of organs) similar in position, structure, and evolutionary origin but not necessarily in function.

26. what is multiplex PCR?

Multiplex polymerase chain reaction (Multiplex PCR) refers to the use of polymerase chain reaction to amplify several different DNA sequences simultaneously (as if performing many separatePCR reactions all together in one reaction). ... In 2008, multiplex-PCR was used for analysis of microsatellites and SNPs.

27. What are plasmids, type, applications, and natural presence?

A plasmid is a small, circular, double-stranded DNA molecule that is distinct from a cell's chromosomal DNA. Plasmids naturally exist in bacterial cells, and they also occur in some eukaryotes. Often, the genes carried in plasmids provide bacteria with genetic advantages, such as antibiotic resistance.

28. What is Relexed Plasmid

These plasmids contain DNA regions called par-locuses (from partition) that control the correct division of the plasmids in the two daughter cells. Plasmids with high copy number are under relaxed control of replication. These plasmids are replicated independently of the replication of the bacterial chromosome.

29. What is selfish DNA?

Selfish genetics elements (also sometimes called selfish DNA) are genetic sequences that spread by forming additional copies of itself within the genome; and makes no specific contribution to the reproductive success of its host organism (it might or might not have significant deleterious effects).

30. What is natural plasmid?

A plasmid is a small DNA molecule within a cell that is physically separated from a chromosomal DNA and can replicate independently. They are most commonly found in bacteria as small circular, double-stranded DNA molecules; however, plasmidsare sometimes present in archaea and eukaryotic organisms.

31.what are the Si RNA importance?

siRNA, which stands for small interfering Ribonucleic Acid, is a class of double-stranded RNA molecules. It is sometimes known as short interfering RNA or silencing RNA.

Small interfering RNA (siRNA) are small pieces of double-stranded (ds) RNA, usually about 21 nucleotides long, with 3' overhangs (2 nucleotides) at each end that can be used to "interfere" with the translation of proteins by binding to and promoting the degradation of messenger RNA (mRNA) at specific sequences.

In doing so, siRNA prevent the production of specific proteins based on the nucleotide sequences of their corresponding mRNA. The process is called **RNA interference (RNAi)**, and may also be referred to as siRNA silencing or siRNA knockdown.

32. what are major or minor grooves ? also describe their size?

The strand backbones are closer together on one side of the helix than on the other. The major groove occurs where the backbones are far apart, the minor groove occurs where they are close together. The grooves twist around the molecule on opposite sides. Certain proteins bind to DNA to alter its structure or to regulate transcription (copying DNA to RNA) or replication (copying DNA to DNA). It is easier for these DNA binding proteins to interact with the bases (the internal parts of the DNA molecule) on the major groove side because the backbones are not in the way.

33. How deoxiribonucleotids join?

They join togeather by phospodiester bond.

34.Write down the steps of cloning strategy. .. 5marks

OVERVIEW OF CLONING STRATEGIES:

Any cell based cloning procedure has 4 essential parts
i). A method of generating DNA fragment for cloning
ii). A reaction that inserts that fragment into the chosen cloning vectors.
iii). A means for introducing that recombinant vector into a host cell wherein it is replicated
iv). A method of selecting recipient cells that have acquired the recombinant

35.Describe disadvantage of Rt-pcr ….. 2 Marks

The **disadvantages of RT-PCR** include its complexity and problems associated with its sensitivity, reproducibility, and specificity. ... However, the introduction of **real-time RT-PCR** and the availability of modern equipment provide extended possibilities for the accurate quantification of mRNA species.

36.Write the use of micro aray in human health….3 Marks

Microbial gene expression analysis

The predominate application of DNA microarray has been to measure gene expression levels.

Applications of microarrays in Profiling in human disease

Arrays have been used to investigate transcriptional profiles associated with human disease and to identify novel disease markers and potential new drug targets.

Genotyping

Microarrays have been widely used as single nucleotide polymorphism (SNP) genotyping platforms.

37.Describe the Principle of si-RNa ….3 Marks

Small interfering RNA (siRNA) is the most commonly used RNA interference (RNAi) tool for inducing short term silencing of protein coding genes. siRNA is a synthetic RNA duplex designed to specifically target a particular mRNA for degradation.

38.Write the Length of major and minor grooves..2 Marks

Major is ~ 22 Angstroms and minor is ~ 12 Angstroms

39. RT-QPCR Disadvantages?

The **disadvantages** of **RT**-PCR include its complexity and problems associated with its sensitivity, reproducibility, and specificity. Moreover, it suffers from the problems inherent in traditional PCR when it is used as a quantitative method (3).

40. How polymerase work?

The DNA polymerases are enzymes that create DNA molecules by assembling nucleotides, the building blocks of DNA. ... When creating DNA, DNA polymerase can add free nucleotides only to the 3' end of the newly forming strand. This results in elongation of the newly forming strand in a 5'-3' direction.

41. Disadvantage of PCR?

Setting up running required high technical skills.

High equipment cost.

High Test Cost

High sterile environment should be provided. (DNA contamination)

42.DNA sequencing its benefits applications and databases? 10 marks

DNA sequencing: Benefits and Applications

DNA sequencing is the process of determining the precise order of nucleotideswithin a DNA molecule. It includes any method or technology that is used to determine the order of the four bases—adenine, guanine, cytosine, and thymine—in a strand of DNA. The advent of rapid
DNA sequencing methods has greatly accelerated biological and medical research and discovery.[1] DNA sequence information is a prerequisite for planning any substantial manipulation of the DNA. Information is useful to various fields i.e. molecular & evolutionary biology, metagenomics, medicine, forensics etc. Techniques for DNA sequencing became available in the late 1970s

Maxam-Gilbert method:

This method for DNA sequencing makes use of chemical reagents to bring about base-specific cleavage of the DNA.

Chain termination or dideoxy procedure

This method was developed by Sanger et al. (1977).It capitalizes on the ability of DNA polymerase to incorporate dideoxynucleoside triphosphates as substrate. DNA synthesis is carried out in the presence of the four deoxynucleoside triphosphates, one or more of which is labelled with 32P and in four separate incubation mixes.

Automated DNA sequencing

To automate the process, it is desirable to acquire sequence data in real time by detecting the DNA bands within the gel during the electrophoretic separation. Automated DNA sequencing offers a number of Advantages.

DNA sequence databases

Since the development of current DNA sequencing technology large amount of sequence data has accumulated that is maintained in 3 data bases

i). National center for Biotechnology Information

ii). DNA Databank of Japan

iii). European Bioinformatics institute-UK

43. Write note on plasmids, its types benefitis, and natural plasmids libraries used? (10 Marks)

Plasmids

- Plasmids are mostly circular DNA molecules that replicate separately from the host chromosome.

- Naturally occurring bacterial plasmids range in size from 5,000 to 400,000 bp
- Plasmid help bacteria to survive in stress.

Phenotypic traits exhibited by plasmids

- Plasmids are widely distributed throughout the prokaryotes, vary in size from less than 1 × 106 to greater than 200 × 106 daltons.
- Plasmids can be categorized into conjugative or non-conjugative depending whether or not they carry a set of transfer genes, called tra genes, which promoted bacterial conjugation.

Natural Plasmids:

Plasmids which were not constructed in vitro for the sole purpose of cloning are called natural plasmids

- Col E1 is a natural occurring plasmid
- RSF2124 is a derivative of Col E1
- pSC101 is natural plasmid from Salmonella panama.

44.What is artifical bacterial chromosome and artifiacl yeast chromosome... 10 marks.

Bacterial artificial chromosomes (BACs):

Bacterial artificial chromosomes are simply plasmids designed for the cloning of very long segments of DNA

(100,000 to 300,000 bp) Bacterial artificial chromosome as cloning vectors

Yeast artificial chromosomes (YACs):

These vectors replicate in yeast like normal chromosome but they have sites where very large DNA fragments can be inserted.

YACs with inserts more than 150,000 bp are nearly as stable as normal cellular chromosomes

Construction of yeast artificial chromosome

45. Protein trafficking: 10

The bacterial inner membrane, periplasmic space and outer membrane all contain proteins not found in the cytoplasm

All of these proteins are first synthesized in the cytoplasm but somehow find their way out

Figure. General secretion pathway

There are actually several general mechanisms of protein export in bacteria

Each system transport a particular type of proteins

Stability of foreign proteins in *E. coli*

Various strategies have been developed to cope with the instability of foreign proteins in *E. coli* In the case of somatostatin, degradation was prevented by producing a fused protein consisting of somatostatin and β-galactosidase

Constructing the optimal promoter

Large No. of promoters for *E. coli* have been analysed Many promoters has led to the formulation of a consensus sequence which consists of the -35 region (5'-TTGACA-) & -10 region or Pribnow box (5'-TATAAT-) Expression from a strong promoter can represent 20-40% of cloned gene product of toral cell protein **Figure.** The base sequence of the -10 and -35 regions for natural and hybrid promoters

46. Define pseudo genes? 2

Pseudogenes are segments of DNA that are related to real genes. Pseudogenes have lost at least some functionality, relative to the complete gene, in cellular gene expression or protein-coding ability.

47. What are the major properties of p1248 in the expression of staphylococcus plasmids? 2

- Expression of *Staphylococcus* plasmid genes in *E. coli*
- pSC101 contain Tetracycline resistance *P1248 from Staphylococcus*

48. Genomic libraries in high capacity vectors? 2

In place of phage-λ derivatives, a number of high capacity cloning vectors such as cosmids, bacterial artificial chromosomes (BACs) and yeast artificial chromosomes (YACs) are available for construction of genomic libraries.

49. Expression cloning in prokaryotes and yeast? 3

50. SNPs as physical markers? 5

Single nucleotide polymorphisms (SNPs) are single base-pair positions in genomic DNA at which different sequence alternatives (alleles) exist in a population SNPs probably are the most important sequence markers for physical mapping of genomes.

Different methods to detect SNPs have been developed that based on enzymatic, electrophoretic, solid phase or chromatographic analysis

51. Write a brief note on proteins primary, secondary, tertiary and quaternary structure? 5

52. Uses of microarray in human health?

The predominate application of DNA microarray has been to measure gene expression levels

Profiling in human disease

- Arrays have been used to investigate transcriptional profiles associated with human disease and to identify novel disease markers and potential new drug targets

Genotyping

Microarrays have been widely used as single nucleotide polymorphism (SNP) genotyping platforms

- M.c.q's.
- 1.microarrays are used to study the ..... of many genes.
- 2.blank from clone-by-clone gene.
- 3.happy mapping.....technique.
- 4.blank from linkage mapping.
- 5.haeIII recognizes..... bp and produce blunt ends.
- 6.psc101 is not advantageous because of.....

7.cosmids have .....site.(cos)

- 8.human genome size is.....
- 9.....is widely used for human genomic libraries.
- 10.drug development and treatmnt....biotechnology.
- 11.theadre schwann proposed an idea which laid foundation of .....
- 12.repeating units in nucleic acids....
- 13.virus is composed of....
- 14.E.coli chromosome have about.....super coils.
- 12.southern blotting is used to detect .....DNA.
- 13.human genome size....
- 14.mRNA is translated into....to perform life activities.
- 14.....degrade a particular transcript of m.RNA in cell to lower expression of particular gene.
- 15.chrmosome walking definition.
- 16.pcr site mutagenesis.
- 17.error frequency when dna lenght is more than 400.

53.